0/1 Knapsack Using LCBB

$M = 12$

W_1	W_2	W_3	W_4	W_5
4	6	3	4	2

P_1	P_2	P_3	P_4	P_5
10	15	6	8	4
[-10	-15	-6	-8	-4]

Convert into max loss.

$m = 5$, — Find P_i / W_i — & put in the decreasing order.

Cost & Upper Bound

without fraction $U = \sum_{i=1}^{m} P_i x_i$

with fraction $C = \sum_{i=1}^{m} P_i x_i$

$\leq m$

$10/4 - 2.5$

$15/6 - 2.5$

$6/3 - 2$

$8/4 - 2$

$4/2 - 2$

If upper bound is lesser we need to update the UB.

(i) First do one random check and fix upper Bound

W_1 W_2, & W_5

$4 + 6 + 2 = 12$.

without fraction

$10 + 15 + 4 = 29$

so $\boxed{U.B = -29 \quad C = -29}$

$UB = \cancel{\infty}$ -29

If cost function for a given node is greater than the upper bound, then the node need not be explored. (Kill that node)

4 10
6 15
3 6

In upper Bound only getting the 2 values

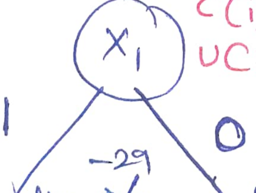
$U = 10 \times 1 + 15 \times 1$
$= -25$

Cost

$C = 10 \times 1 + 15 \times 1 + 2/3 \times 6$
$= -29$

$C(2) = -29$
$U(2) = -29$

C (min) choose



$C(1) = -29$
$U(1) = -29$

$U = -\cancel{25}$ -29
$C = -29$

$U = -21.25$
$C = -27$

$C(1,2,5)$ $10 \times 1 + 15 \times 1 + 4 \times 1$
$= (-29)$

$C(3) = -27$
$U(3) = -25$

X

15, 1
6, 3
21

Cost
$= 15 \times 1 + 6 \times 1 + 3/4 (8)$
$= -27$
$-15 - 6 - 4$
$= -25$

This side x_2 is taken mandatory

$x_1 = 1$

x_2 is not taken mandatory

$x_2 = 1$ (1)

$x_2 = 0$ 0

Upper Bound $\frac{4}{10} + \frac{6}{15} + \frac{2}{4} = -29$

UB = $\frac{4}{10} + \frac{3}{6} + \frac{4}{8} = -24$

Cost $\frac{4}{10} + \frac{6}{15} + \frac{2/3}{6} = -29$

min

Cost = $\frac{4}{10} + \frac{3}{6} + \frac{4}{8} + \frac{1}{2}(4) = -26$

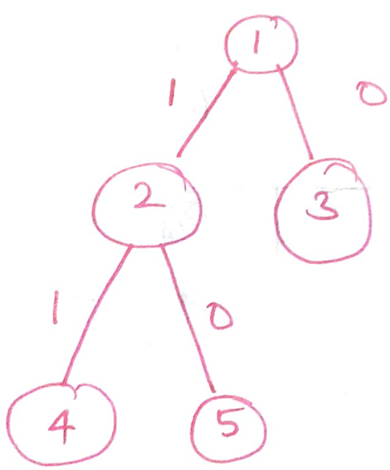
X

$u(4) = -29$

$c(4) = -29$

$u(5) = -24$

$c(5) = -26$



Similarly $x_3 = 1$ is taken Mandatory

$x_3 = 1$

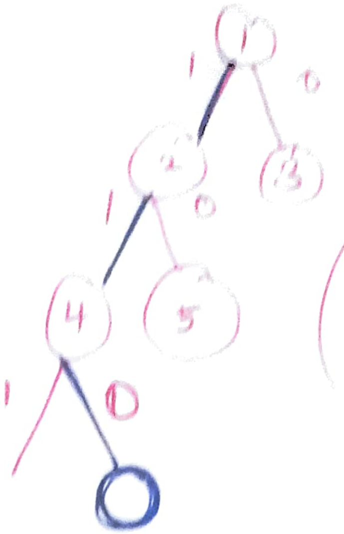
$u(6) = -10 - 15 \cancel{-4} = -25$

$c(6) = -10 - 15 - \left(\frac{2}{\cancel{3}}\right)\cancel{6}^{2} = -29$

$x_3 = 0$

$u(7) = -10 - 15 - 4 = -29$

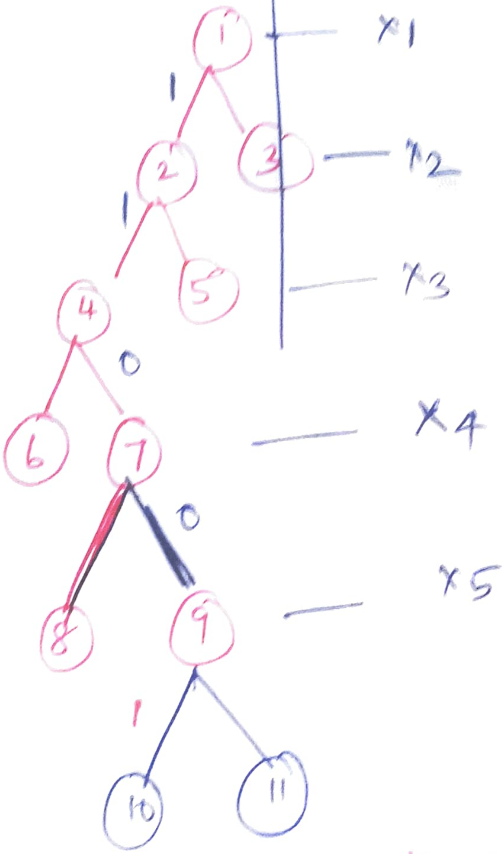
$c(7) = -10 - 15 - \left(\frac{2}{\cancel{4}}\right)\cancel{8}^{2} = -29$



$x_1 = 1$
$x_2 = 1$
$x_3 = 0$

$x_4 = 1$	$x_4 = 0$
$u(8) = -10-15 = -25$	$u(9) = -10-15-4 = -29$
$c(8) = -10-15-\frac{2}{4}(8) = -29$	$c(9) = -10-15-4 = -29$

x



$x_1\ x_2\ x_3\ x_4$
$(1, 1, 0, 0)$

$x_5 = 1$	$x_5 = 0$
$u(10) = -10-15-4 = -29$	$u(11) = -10-15-0 = -25$
$c(10) = -10-15-4 = -29$	$c(11) = -10-15-0 = -25$

$(1, 1, 0, 0, 1) = (1, 1, 0, 0, 1) = 29$